School of Information and Physical Sciences
Process Analysis and Problem Solving
INFO6050

Assessment 1	: Last-Mile Delivery Route Optimisation
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Lecture Timings	: Monday 3.00 PM – 5.00 PM
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Figure 1: Illustration of last-mile delivery challenges in Australian urban and rural areas [7]

Executive Summary:

This report looks closely at how Australian companies like Australia Post, PFD Food Services, and Coca-Cola Euro pacific Partners Australia handle their last-mile delivery

routes today. These companies use smart tools like AI, live tracking, and flexible routing to deal with busy city streets and wide rural areas. We include an activity diagram to show Australia Post's step-by-step process, including key decisions they make along the way. Safety is a big focus, with plans to check for weather dangers, theft risks, and driver health, such as heat problems and using tracking systems to keep everyone safe.

The report proposes several initiatives, including enhancing AI, testing out drones, forming relationships with local communities, incorporating IoT devices, and utilizing smart lockers for deliveries, to advance delivery efficiency and ESG objectives. The initiatives can lead to cost savings, lower carbon footprint, and faster deliveries, thus enhancing organizational resilience and sustainability.

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1. Introduction:

I am a trainee business analyst at a big logistics company. This report explores ways to improve our last-mile delivery and help our environmental and social goals, called ESG. Last mile delivery is the final step where goods move from a warehouse to a customer's door. It often costs a lot because of fuel, traffic jams, and last minute changes [1].

Australia has many different places that make delivery tricky. Big cities like Sydney, Melbourne, and Brisbane have lots of cars, narrow roads, and many people, which makes finding the best route hard [2]. Faraway rural areas, like the outback or remote coasts, have long distances, rough roads, and changing weather like floods or heatwaves, so we need special plans to handle these challenges [3].

We use eight special ways from Yunkaporta (2009) to guide this report for my company. First, non-linear thinking happens as I prepare the report and then rethink what I've learned, going back to adjust my ideas based on new information. Second, non-verbal learning comes from writing this report, thinking deeply, and doing lots of research to

understand the problem. Third, deconstruct and reconstruct means I look at the big picture of delivery challenges, then break them into smaller parts like route planning, and put them back together with better solutions. Fourth, storytelling is how I present this report, sharing a clear story about how deliveries work and how we can improve them.

Fifth, community connects to the Australian companies I chose—Australia Post, PFD Food Services, and Coca-Cola—and also to the wider business analyst world where we share ideas. Sixth, land links focus on Australia's unique places, like flood-prone Queensland or dry Western Australia, which affect how we plan routes and keep things safe [6]. Seventh, learning maps involve creating visual guides, like activity diagrams, to show the delivery process clearly. Eighth, symbols are used in these diagrams and pictures, like diamonds for decisions or arrows for flow, to make complex ideas simple [10].

The land in Australia plays a big role. For example, flood areas in Queensland or dry zones in Western Australia affect how safe and green our deliveries can be [6]. We use pictures, diagrams, and lots of research to tell this story. We also think creatively by checking data, processes, and risks over and over again to find new ideas [10]. Diagrams, like activity maps, help us see the delivery process clearly, and simple symbols in these diagrams show how everything works together easily [4].

2. Current Examples:

Last-mile delivery means using data to find the quickest and cheapest routes, which saves time, fuel, and money [1]. In Australia, cities deal with traffic and rivers, while rural areas face few roads and tough weather [6]. Companies use AI to change routes based on live traffic and delivery schedules [7]. These examples show how they use non-linear thinking to adjust plans, non-verbal learning through research, and community ties to improve services, all shaped by Australia's land links.

2.1 Australia Post:

Australia Post uses e2open software to plan their routes and give accurate delivery times [2, 12]. They start with orders, pick up parcels, and deliver them to homes. They use GPS and smart routing to handle city traffic or rural challenges [9]. In cities, they group postcodes and avoid jams, in rural areas, they use satellite maps to save up to 15% on fuel [8]. They also work with local post offices to drop off parcels, building community trust. During floods in Queensland, they adjust routes using real time weather data, showing non-linear thinking. They face challenges like busy holiday seasons, so they add extra drivers and use apps to track packages, ensuring customers know when deliveries arrive. Their storytelling shines through clear updates to customers, and land links are key with Australia's varied area.

Australia Post also tests electric vans in cities like Melbourne to cut emissions, aligning with ESG goals [12]. They train drivers to handle heatwaves and wildlife, like kangaroos on rural roads, using symbols in training guides. Their learning maps include detailed route plans, and they deconstruct delivery problems by analyzing traffic patterns, then reconstruct better schedules. Community efforts include partnering with small towns for pickup points, making deliveries easier for remote areas [12].

2.2 PFD Food Services:

PFD Food Services, a company that delivers food, uses Complexica's Decision Cloud to plan and send out fresh items [13]. Their AI finds good routes, considering traffic, weather, and delivery times, especially between cities and rural spots [3]. In cities, they fix delays fast, in rural areas, they use local depots and test drones to keep food fresh and cut waste [3]. They work with local farmers to source goods, strengthening community ties. When storms hit, they use weather apps to reroute trucks, showing nonlinear adjustments.

Struggles include preserving food temperature when it is hot, using refrigerated trucks and checking on them frequently [13]. PFD utilizes IoT sensors to monitor temperature and location to keep things safe. Their storytelling comes through happy customer reviews about fresh deliveries. Land links are important, as they adapt to rugged outback roads and coastal floods [6]. They deconstruct delivery issues by studying demand patterns, then reconstruct plans with better timing.

PFD is trying electric delivery vans to reduce pollution and works with rural shops as pickup spots. They train drivers to avoid animals like emus and use learning maps to plan efficient routes. Symbols in their apps, like green dots for safe zones, guide drivers. Community efforts include donating extra food to local charities, showing their commitment to the area [13].

2.3 Coca-Cola Europacific Partners Australia:

Coca-Cola uses Adiona software to plan routes and manage their delivery trucks [11]. They balance truck loads, check traffic, and use simulations to save money in cities and rural areas [8]. In cities, drivers use apps to adjust plans on the spot, in rural areas, they plan for long trips to lower emissions [9]. They partner with local stores to stock shelves, building community connections. During bushfires, they reroute using live maps, reflecting non-linear thinking.

They experience issues such as the heavy loads in the cities and long distances in the bush. Coca-Cola employs information systems through telematics to track truck speeds and fuel usage, and they use the information to reduce costs [11]. Their storytelling is evident in advertisements featuring satisfied customers enjoying their

cold drinks. Land links matter, as they adjust for Australia's heat and remote roads. They deconstruct delivery by analyzing customer demand, then reconstruct with optimized schedules.

Coca-Cola tests hybrid trucks to save fuel and trains drivers for safety in dust storms [11]. Learning maps help plan routes, and symbols like red flags in apps warn of hazards. Community efforts include sponsoring local sports teams, and they use non-verbal methods like detailed reports to improve processes. Their focus on safety and efficiency ties into ESG goals.

These examples show how companies share ideas to make deliveries better across Australia's different places.

Table 1: Process comparison: (Compiled from [1], [2], [3], [8], [11], [12], [13], [14])

Company	Urban Process	Rural Process	Technology Used
Australia Post	Sequencing, ETAs via e2open	Geographic adaptation	GPS, AI
PFD	Dynamic weather adjustments	Depot integration	Decision Cloud
Coca-Cola	Contractor stabilisation	Simulation-based planning	Adiona, Simio

3. Activity Diagram:

This activity diagram explains how Australia Post manages last-mile delivery. It has three roles: System, Dispatcher, and Driver. The diagram breaks the process into clear steps, uses diamond shapes for choices, and arrows to show the flow, making it easy to understand [4]. This is a learning map that visually guides us through the process.

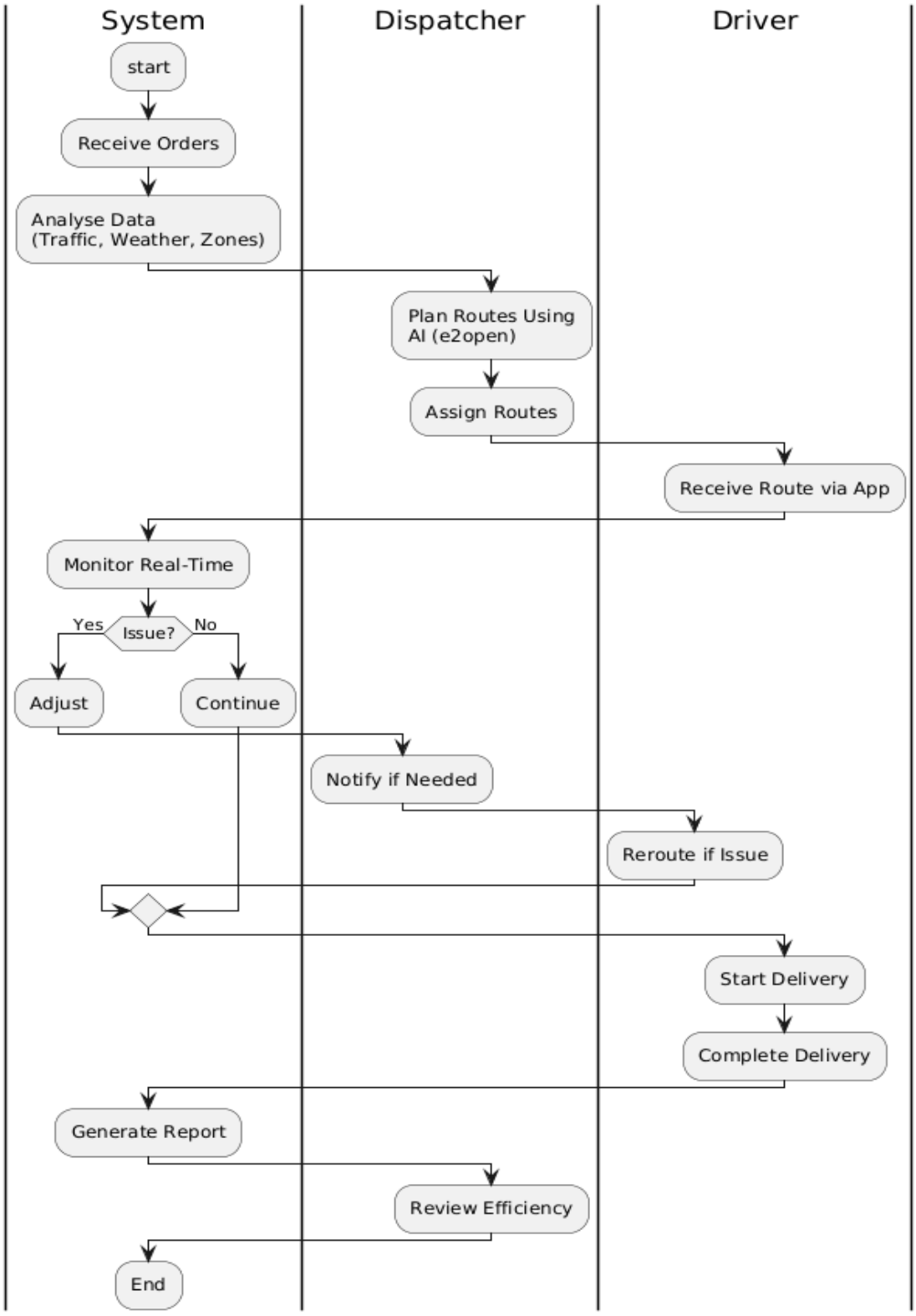


Figure 2: Australia POST Activity diagram showing the process flow with roles.

The diagram starts with the System getting orders and checking traffic, weather, and zones. The Dispatcher then uses e2open to plan routes and assigns them to drivers. The

Driver gets the route on an app and begins delivery. The System watches everything live, if there's a problem like a traffic jam, it changes the route, and the Dispatcher tells the Driver to take a new path. If no problem, delivery goes on as planned. The Driver finishes the job, the System makes a report, and the Dispatcher checks how well it went. Symbols like diamonds and arrows make the diagram clear, and non verbal learning comes from studying it.

4. Safety Considerations:

Making delivery routes better means keeping drivers, trucks, and goods safe. We look at risks like theft, damage, and driver health, using technology and rules that fit Australia's conditions [3]. This involves deconstructing safety issues and reconstructing safer plans.

In cities, Australia Post and Coca-Cola use live monitoring and AI to stay away from risky spots, like areas with theft or heavy traffic that could cause accidents. They use geo-fencing to keep trucks on safe roads, and apps warn about road closures [9, 8]. Land links are important here with city layouts.

In rural areas, PFD and Australia Post use weather apps to avoid floods or bushfires, which happen a lot in Queensland or the Northern Territory. They also check trucks regularly to stop breakdowns where help is hard to find, keeping drivers safe [3, 6]. This shows non-linear adjustments to weather changes.

Driver safety is very important, especially in Australia's hot weather. Heat can make electric vans lose power and make drivers feel tired or sick. Apps track driving hours and suggest breaks, while Coca-Cola trains drivers to handle dust storms or animals on the road [2, 9]. Community support comes from training programs.

All companies use telematics to stop speeding, which is a big safety worry because of tight delivery times, and they have emergency plans for accidents. They plan safe parking spots and track drivers to follow rules. These steps help reduce crashes and support ESG goals [7, 5]. Storytelling here shares safety success stories.

5. Recommendations:

Based on what I learned, here are few detailed suggestions:

1. Use AI to predict routes better, like Australia Post does, with IoT sensors giving real-time updates to fix delays in cities and problems in rural areas [7]. This uses non-linear thinking.
2. Test drones for rural deliveries to reach far places safely and quickly. In Australia, this can work with government rules for small packages [3]. Land links guide drone use.
3. Switch to electric vans to cut pollution and save fuel, making deliveries greener [3]. This supports ESG goals.

4. Add IoT devices to track goods and routes live, helping drivers adjust plans on the go [7]. Learning maps can show this process.
5. Set up smart lockers in cities where customers can pick up parcels anytime, reducing missed deliveries [6]. Community input helps place them.
6. Work with local communities for road advice, partnering with rural groups and adding lockers to build trust and safety [6]. This strengthens community ties.
7. Improve route planning with better software that changes paths based on weather or traffic [9]. Symbols in apps guide drivers.
8. Train drivers more on staying safe in heat, storms, or with animals, keeping them healthy [2]. Non-verbal training materials help.

These ideas use new tech like self-driving vans and cloud systems, which could save 20-30% on costs and help ESG by lowering emissions and improving safety [3, 9]. Deconstruct and reconstruct methods shape these plans.

6. Conclusion:

This report looks at last mile delivery problems, rebuilds the process with examples and diagrams, and suggests ways to improve ESG. By studying Australian companies and their landscapes, it shows how to deliver goods efficiently and sustainably. This encourages new and creative ideas in the business analyst world [10]. Storytelling ties it all together, with nonlinear thinking and symbols enhancing the message.

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